

APPOINTMENTS **École normale supérieure, Paris School of Economics** Paris, France
Postdoctoral Researcher (incoming) Oct 2026 – Sep 2028
 Supervisor: Prof. Matthew Gordon

EDUCATION **University of Illinois at Urbana-Champaign** Champaign, IL, USA
Ph.D. in Economics Aug 2020 - Aug 2026
 Advisor: Prof. Tatyana Deryugina

University of Illinois at Urbana-Champaign Champaign, IL, USA
M.S. in Policy Economics Dec 2019

University of International Business and Economics (UIBE) Beijing, China
B.S. in Economics (Minor in French) Jun 2018

University of California, Berkeley Berkeley, CA, USA
Exchange student Jan 2017 - May 2017

RESEARCH INTERESTS **Environmental Economics, Public Economics, Applied Econometrics**

WORKING PAPERS **Air Pollution and Economic Activity: Evidence from Foot Traffic Patterns (Job Market Paper)**

I investigate how air pollution affects economic activity. Using over 37 billion phone-location-based foot traffic data points from SafeGraph, I conduct a large-scale analysis to examine the causal effect of air pollution on activity patterns across the U.S. Using changes in local wind direction as an instrumental variable for air pollution, I characterize the dynamic response to pollution exposure. I find that PM2.5 reduces economic activity in both the short and medium run, with effects persisting for up to two weeks before gradually recovering. The reductions are widespread across different economic sectors, with recreational activities experiencing the largest decline. The effect is more pronounced in higher-income counties and areas with larger shares of children, suggesting greater awareness among wealthier or more vulnerable populations.

Go with the Wind: Polluters' Strategic Response to Wind Directions

I investigate the strategic behavior of polluters in the U.S. Under the Clean Air Act, air quality monitor data determine whether an area meets environmental standards. Because wind carries pollution, monitors detect more pollution from upwind sources and less from downwind sources. This disparity incentivizes polluters to emit less when they are upwind of a monitor and more when they are downwind. I identify such strategic behavior among U.S. coal-fired power plants, finding that a one-standard-deviation increase in favorable wind direction—when wind blows pollutants away from monitors—leads to a 0.8% (172 lbs) increase in sulfur dioxide (SO₂) emissions and a 0.4% (54 lbs) increase in nitrogen oxides (NO_x) emissions. At the same time, fuel input remains constant while emission rates rise, suggesting that power plants temporarily turn off pollution-control equipment. The increase is more pronounced when power plants are in a nonattainment county, located in the same state as the monitor, or surrounded by fewer nearby polluters. These findings suggest that polluters facing stricter regulatory pressure are more likely to respond strategically when conditions are favorable.

Information and Climate Adaptation: Evidence from ENSO (with Tatyana Deryugina and Alan Holowinski; draft available upon request)

We study whether early climate information affects economic adaptation. Using data from Peru, we examine how farmers respond to early-season signals of the El Niño–Southern Oscillation (ENSO), which provide imperfect forecasts of growing-season weather. Exploiting spatial variation in the strength of ENSO teleconnections, we show that farmers adjust crop allocations more in locations where ENSO better predicts local conditions and in crops more sensitive to weather. We further find that realized weather shocks are less damaging when anticipated by early signals, while forecast errors generate yield losses. These results show that climate information shapes production decisions and mediates the economic impacts of climate variability.

WORK IN PROGRESS

Natural Disasters and Occupational Mobility: Evidence from the 1927 Mississippi Flood (with Xiaohan Wang; draft available upon request)

We investigate how a major natural disaster affects intergenerational occupational mobility in the agricultural sector. Linking individuals across the 1920, 1930, and 1940 Censuses, we show that children residing in counties affected by the 1927 Mississippi Flood were significantly less likely to become farmers as adults. This effect is mainly driven by children from non-farming families, who were 9% less likely to work in agriculture and 10% more likely to transition into the service sector. In the longer run, these individuals achieved socioeconomic status scores about 3% higher than comparable children in non-flooded counties. Our findings suggest that natural disasters can accelerate structural change by shifting new labor market entrants out of agriculture and into higher-status occupations.

Social Networks and Household Protective Investments: Evidence from Wildfire Smoke (with Sanjukta Mitra)

Deforestation and Air Pollution in the Indonesian Rainforest (with Francesco Cenerini)

CONFERENCES AND WORKSHOPS

2026

ASSA Annual Meeting (Philadelphia), Annual Conference of the American Society of Health Economist (Minneapolis), Canadian Economics Association Annual Conference (invited presentation, Vancouver)

2025

Canadian Resource and Environmental Economics Association Annual Conference (Winnipeg, Canada), Southern Economic Association Annual Conference (Tampa), Heartland Workshop (poster, UIUC), Program in Environmental and Resource Economics Seminar (UIUC), Midwest Economics Association Annual Conference (Kansas City), Eastern Economic Association Annual Conference (New York City), Applied Micro Research Lunch (UIUC), Graduate Student Research Seminar (UIUC)

2024

Occasional Workshop in Environmental and Resource Economics (UCSB), Online Summer Workshop in Environment, Energy, and Transportation (Virtual), Applied Micro Research Lunch (UIUC), Graduate Student Research Seminar (UIUC)

2023

Price Theory Summer Camp (University of Chicago), Applied Micro Research Lunch (UIUC)

2022

Berkeley/Solan Summer School in Environmental and Energy Economics, Applied Micro Research Lunch (UIUC)

TEACHING EXPERIENCE	ECON 102: Microeconomic Principles	Fall 2021, Spring 2022
	Teaching Assistant	
	The Economics of the Firm (EMBA at the University of Warsaw)	Fall 2019
	Teaching Assistant to Prof. Hadi Esfahani	
	BADM 529: Macroeconomics	Summer 2019
	Course Assistant	
	ECON 528: Microeconomics for Business	Spring 2019, Summer 2019
	Course Assistant	
WORK EXPERIENCE	Gies College of Business, UIUC	
	Research Assistant to Prof. Tatyana Deryugina	Aug 2022 –present
	Department of Economics, UIUC	
	Research Assistant to Prof. Eunyi Chung	June 2021 –Aug 2022
	China Institute for Educational Finance Research, Peking University	
	Research Associate	Mar 2020 –Aug 2020
	Risk Management Department, Bank of Communications	
	Data Analyst Intern	Jun 2017 –Aug 2017
	School of International Trade and Economics, UIBE	
	Research Assistant to Prof. Deyu Yuan	Sep 2016 –Jan 2017
AWARDS	• Robert Willis Harbeson Memorial Dissertation Fellowship, University of Illinois	2025
	• Hung-Chao and Julia Tai Family Fellowship, University of Illinois	2025
	• Conference Travel Fellowship, University of Illinois	2025
	• Cleo Fitzsimmons Award, University of Illinois	2022
	• Graduate Fellowship, University of Illinois	2020
ACADEMIC SERVICES	Referee for <i>Journal of Environmental Economics and Management</i> ; <i>Quarterly Review of Economics and Finance</i>	
SKILLS	Languages: Chinese (Native), English (Fluent), French (Basic).	
	Programming: R, Python, Stata, Git, Shell.	

REFERENCES

Tatyana Deryugina (Chair)

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